Justin Toh, Ph.D.

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RESEARCH EXPERIENCE

Postdoctoral Scientist, Advisor: Hyundae Cho, PhD *2023-2025*
CrossLife Technologies, Carlsbad, CA

- Led T7 phage display-guided directed evolution with the patented GyrI-like antibody-mimetic scaffold; discovered high-affinity binders to SARS-CoV-2 spike protein, fentanyl, HER2, and Alzheimer's disease tau
- Produced and purified recombinant proteins via bacterial expression and ÄKTA Start affinity chromatography
- Analyzed AlphaFold-predicted models and experimentally determined crystal structures in PyMOL to map putative binding epitopes and guide library design
- Measured kinetics (K_D, k_{on}, k_{off}) by SPR on OpenSPR and quantified apparent affinity (EC_{50}) by ELISA
- Mentored 1 research associate and 2 postdoctoral scientists to execute SOP-driven workflow
- Led monthly journal club; aligned literature to project goals, informing assay design and troubleshooting

Graduate Research Fellow, Advisor: Christian Tschudi, PhD *2015-2023*
Yale University, Department of Microbial Pathogenesis, New Haven, CT

- Identified 22 genes regulating virulence and developmental progression in *Trypanosoma brucei* via RNAi screening
- Discovered 4 RNA-binding proteins that trigger infective-stage differentiation within 24 hours (normally 3-5 weeks)
- Mapped transcriptomic changes (RNA-Seq) and RNA-binding protein targeted mRNAs (seCLIP) with DNASTAR
- Analyzed the secondary structure of *T. brucei* vault RNA using an RNase H cleavage assay, contributing to the discovery of its role in trans-splicing
- Collaborated on a cross-disciplinary team to design and prototype 3D-printed fixtures that help isolate aberrant, adhesive parasites for downstream analysis
- Evaluated small molecule α-PanAm drug efficacy against drug-sensitive and drug-resistant *Plasmodium falciparum in vitro* and *in vivo* (Choukri Ben Mamoun laboratory)

Undergraduate Research Scholar, Advisor: Peter J. Bradley, PhD *2011-2015*
UCLA, Department of Microbiology, Immunology, and Molecular Genetics, Los Angeles, CA

- Adapted CRISPR/Cas9 and BioID, an approach for studying protein interactions, in *Toxoplasma gondii*
- Discovered, molecularly cloned, characterized, and named the IMC Suture Components (ISC)

SKILLS

PROTEIN ENGINEERING & ASSAYS: T7 phage display–guided directed evolution, recombinant protein expression & purification (E. coli, ÄKTA Start), Surface Plasmon Resonance (OpenSPR), ELISA, Western blot; immunofluorescence

MOLECULAR BIOLOGY: DNA manipulation & subcloning, PCR, qRT-PCR, isothermal amplification (LAMP, RPA, LFA readout), site-directed mutagenesis, CRISPR/Cas9 (knock-in, knock-out), transfection, flow cytometry, parasite and mammalian cell culture, proximity labeling (BioID), GraphPad Prism

GENOMICS & RNA BIOLOGY: Transcriptomic analysis (RNA-Seq, seCLIP), ChIP-seq, polysome profiling, RNAi knockdown screening, Northern blot, RNase H assay, EMSA assay, DNASTAR

EDUCATION

Yale University, New Haven, CT
Doctor of Philosophy, Microbial Pathogenesis *2023*

Dissertation: Identification and characterization of regulators in the developmental process leading to infectious *Trypanosoma brucei*

University of California Los Angeles, Los Angeles, CA
Bachelor of Science, Microbiology, Immunology, and Molecular Genetics, magna cum laude (GPA: 3.87) *2015*

TEACHING

Graduate Teaching Fellow, Yale Graduate School of Arts & Sciences *2018-2019, 2022*
Yale University, New Haven, CT

- Lectured as a guest speaker in MCDB105 at Yale about the relevance of parasitic organisms in modern society
- Instructed two semesters of microbiology techniques in MCDB 291L and received 100% positive reviews

PUBLICATIONS

Toh JY, Nkouawa A, Dong G, Kolev NG, Tschudi C. (2023). Two cold shock domain containing proteins trigger the development of infectious *Trypanosoma brucei. PLOS Pathogen.*

Toh JY, Nkouawa A, Sánchez SR, Shi H, Kolev NG, Tschudi C. (2021). Identification of positive and negative regulators in the stepwise developmental progression towards infectivity in *Trypanosoma brucei*. *Sci Rep.*

Kolev NG, Rajan KS, Tycowski KT, **Toh JY**, Shi H, Lei Y, Michaeli S, Tschudi C. (2019). The vault RNA of *Trypanosoma brucei* plays a role in the production of trans-spliced mRNA. *J Biol Chem.*

Chiu JE, Thekkiniath J, Choi JY, Perrin BA, Lawres L, Plummer M, Virji AZ, Abraham A, **Toh JY**, Zandt MV, Aly ASI, Voelker DR, Mamoun CB. (2017). The antimalarial activity of the pantothenamide α-PanAm is via inhibition of pantothenate phosphorylation. *Sci Rep.*

Chen AL, Moon AS, Bell HN, Huang AS, Vashisht AA, **Toh JY**, Lin AH, Nadipuram SM, Kim EW, Choi CP, Wohlschlegel JA, Bradley PJ. (2017). Novel insights into the composition and function of the *Toxoplasma* IMC sutures. *Cell Microbiol.*

Choi JY, Kumar V, Pachikara N, Garg A, Lawres L, **Toh JY**, Voelker DR, and Mamoun CB. (2016). Characterization of Plasmodium phosphatidylserine decarboxylase expressed in yeast and application for inhibitor screening. *Mol. Microbiol.*

Chen AL, Kim EW, **Toh JY**, Vashisht AA, Rashoff AQ, Van C, Huang AS, Moon AS, Bell HN, Bentolila LA, Wohlschlegel JA, Bradley PJ. (2015). Novel components of the *Toxoplasma* inner membrane complex revealed by BioID. *mBio*.

SELECTED CONFERENCES

8th Kinetoplastid Molecular Cell Biology Meeting, *Woods Hole, MA* *27 April-1 May 2019*

Poster: A targeted RNAi screen identifies genes that play a role at different stages of *Trypanosoma brucei* metacyclogenesis

Speaker: Two cold-shock domain proteins in *Trypanosoma brucei* are involved in the developmental regulation of metacyclic VSG genes

Amgen Scholars Poster Session, *Los Angeles, CA* *26 August 2014*

Poster: Characterization of novel *Toxoplasma* Inner Membrane Complex proteins identified from a BioID approach

Molecular Parasitology Meeting, *Woods Hole, MA* *August 2013*

Poster: Structure and function of the *Toxoplasma* ISPs: a family of conserved Inner Membrane Complex proteins

SELECTED HONORS & AWARDS

Representative "RNA" article for the 2019 JBC retrospective collection	*2020*
Multidisciplinary Parasitology Training Program (T32AI007404)	*2016-2018*
Amgen Scholars Program	*2014*
Undergraduate Research Scholar Program	*2014-2015*
Southern California Fukienese Association Scholarship	*2011-2015*